

Pranjal Naman

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EDUCATION

Indian Institute of Science (IISc)

Ph.D. Engineering in Computational & Data Sciences; **CGPA: 9.4/10**

Thesis: *Scalable Abstractions and Platforms for GNN Training and Inference on Large-Scale Dynamic Graphs*

Netaji Subhas Institute of Technology (NSIT), University of Delhi

B.E. in Instrumentation & Control Engineering; **CGPA: 9.14/10**

Bengaluru, India

Aug 2022–Current

New Delhi, India

Aug 2016–Jul 2020

RESEARCH EXPERIENCE

PhD Student, DREAM Lab, Indian Institute of Science

(Principal Investigator: Prof. Yogesh Simmhan)

Design and build scalable system architectures and abstractions for efficient GNN training and inference on large, dynamic, and distributed graph datasets.

Co-optimize graph characteristics, model design, and distributed hardware to enable low-latency, high-throughput, and practical GNN deployment at scale.

CONFERENCE/JOURNAL PUBLICATIONS

- [1] **Pranjal Naman** and Yogesh Simmhan. Atlas: Efficient out-of-core inference for billion-scale graph neural networks. In *35th ACM International Symposium on High-Performance Parallel and Distributed Computing (HPDC)*, 2026. Paper  Arxiv .
- [2] **Pranjal Naman** and Yogesh Simmhan. Optimes: Optimizing federated learning using remote embeddings for graph neural networks. *Journal of Parallel and Distributed Computing (JPDC)*, 2026. Paper  Arxiv .
- [3] Bharadwaj Dasari, Turaga Sai Dhiraj, Ganesh Jambhrunkar, Thirumalai Kailasam, Charu Vikram, Saurav Singla, **Pranjal Naman**, and Yogesh Simmhan. Billion-scale fintech analytics: Scalable data management and anomaly detection at npc. In *42nd IEEE International Conference on Data Engineering (ICDE)*. IEEE, 2026. Paper .
- [4] **Pranjal Naman** and Yogesh Simmhan. Ripple: Scalable incremental gnn inferencing on large streaming graphs. In *2025 IEEE 45th International Conference on Distributed Computing Systems (ICDCS)*, pages 857–867. IEEE, 2025. Paper  Arxiv .
- [5] **Pranjal Naman** and Yogesh Simmhan. Optimizing federated learning using remote embeddings for graph neural networks. In *European Conference on Parallel Processing (EuroPar)*, pages 470–484. Springer, 2024. Paper  Arxiv .

WORKSHOP PUBLICATIONS

- [1] Hrishikesh Haritas, **Pranjal Naman**, Mohit Agarwal, Saurav Singla, and Yogesh Simmhan. Computing temporal graph centrality measures over billion-scale financial networks. In *9th Joint Workshop on Graph Data Management Experiences & Systems (GRADES) and Network Data Analytics (NDA), colocated with SIGMOD/PODS (To Appear)*, 2026.
- [2] Akash Sharma, **Pranjal Naman**, Roopkatha Banerjee, Priyanshu Pansari, Sankalp Gawali, Mayank Arya, Sharath Chandra, Arun Josephraj, Rakshit Ramesh, Punit Rathore, et al. Scaling real-time traffic analytics on edge-cloud fabrics for city-scale camera networks. *26th IEEE International Symposium on Cluster, Cloud, and Internet Computing Workshops (CCGridW SCALE Challenge)*, 2026. **Winner, CCGrid TCSC SCALE Challenge 2026** Paper  Arxiv .
- [3] **Pranjal Naman** and Yogesh Simmhan. A gpu is all you need: Rethinking distributed and out-of-core gnn training. In *2025 IEEE 32nd International Conference on High Performance Computing, Data and Analytics Workshop (HiPCW)*. IEEE, 2025. **Best Student Poster and Lightning Talk**. Paper .
- [4] Hrishikesh Haritas, Abhinav Rawat, **Pranjal Naman**, Ganesh Jambhrunkar, Amit Khandelwal, Saurav Singla, and Yogesh Simmhan. Towards scalable mining of temporal graph motifs over large-scale transaction networks. In *2025 IEEE 32nd International Conference on High Performance Computing, Data and Analytics Workshop (HiPCW)*. IEEE, 2025. Paper .
- [5] **Pranjal Naman** and Yogesh Simmhan. Performance trade-offs in gnn inference: An early study on hardware and sampling configurations. In *2024 IEEE 31st International Conference on High Performance Computing, Data and Analytics Workshop (HiPCW)*, pages 173–174. IEEE, 2024. Paper .
- [6] **Pranjal Naman** and Yogesh Simmhan. Topology-aware aggregation for federated graph learning. In *European Conference on Parallel Processing Workshops (EuroParW)*, pages 417–422. Springer, 2024. Paper .

- [7] **Pranjal Naman**, Suved Sanjay Ghanmode, and Yogesh Simmhan. Evaluating strategies for federated graph learning. In *2023 IEEE 30th International Conference on High Performance Computing, Data and Analytics Workshop (HiPCW)*. IEEE, 2023. Paper [↗](#) .
- [8] **Pranjal Naman** and Yogesh Simmhan. Performance modelling of graph neural networks. In *2023 IEEE/ACM 23rd International Symposium on Cluster, Cloud and Internet Computing Workshops (CCGridW)*, pages 334–336. IEEE, 2023. Paper [↗](#) .
- [9] Varad Kulkarni, Akarsh Chaturvedi, **Pranjal Naman**, and Yogesh Simmhan. To think like a vertex (or not) for distributed training of graph neural networks. In *2023 IEEE/ACM 23rd International Symposium on Cluster, Cloud and Internet Computing Workshops (CCGridW)*, pages 351–353. IEEE, 2023. Paper [↗](#) .

PREPRINTS

- [1] **Pranjal Naman** and Yogesh Simmhan. Taurus: Accelerating out-of-core graph neural network inference on billion-scale graphs. *Under review*, 2026. Arxiv [↗](#) .
- [2] **Pranjal Naman**, Parv Agarwal, Hrishikesh Haritas, and Yogesh Simmhan. Ripple++: An incremental framework for efficient gnn inference on evolving graphs. *Under review*, 2026. Arxiv [↗](#) .

AWARDS AND SCHOLARSHIPS

- Winner, CCGrid TCSC SCALE Challenge, CCGrid 2026, Sydney, Australia.
- SIGMOD/PODS India Fellowship to attend SIGMOD/PODS 2026, Bengaluru, India.
- Best Student Research Poster and Lightning Talk, 32nd International Conference on High Performance Computing, Data and Analytics, 2025, India.
- ACM India-IARCS Travel Grant to present my work at ICDCS 2025, Glasgow.
- Young Researcher, Pingala Interactions in Computing, 2025.
- HiPC Student Research Symposium Travel Grant in 2023, 2024, and 2025.
- Euro-Par PhD Studentship to present my work at EuroPar 2024, Spain.
- Full Tuition Waivers for being in top 5% students, all 4 years 2016–20, B.E. in Instrumentation & Control Engg., Netaji Subhas Institute of Technology, Delhi.

PROFESSIONAL SERVICE

- Lead Volunteer at 2026 ACM SIGMOD/PODS Conference, Bengaluru, India.
- Reviewer, Applied Data Science Track, 13th ACM IKDD International Conference on Data Science (CODS), 2025.
- Shadow Program Committee Member, ACM SIGMETRICS, 2026.
- Teaching Assistant, DS256: Scalable Systems for Data Science, Indian Institute of Science, Jan 2026–Present
- Volunteer at 2023 IEEE/ACM 23rd International Symposium on Cluster, Cloud and Internet Computing Workshops

EMPLOYMENT

- Software Engineer, Soroco, Bengaluru, India, Sep 2020–Jul 2021
- Software Engineer Intern, Nucleus Software, Noida, India, May 2019–Jul 2019